New 0inflated models: Poisson & Binomial

Parametrisation

This is a new implementation (Nov'22) of zero-inflated Poisson, Binomial and negative Binomial likelihood, where we will allow for a linear predictor in both the zero-inflattion and in the mean, but one of them needs to consists of fixed effects only. This means the setup will be somewhat different that for other likelihood models.

Details

The zero-inflated likelihood $f_0(y|\dots)$ is defined as

$$f_0(y|\eta_1, \eta_2) = p(\eta_1)1_{[y=0]} + (1 - p(\eta_1))f(y|\eta_2)$$

where $f(y|\dots)$ is either Poisson, Binomial or negative Binomial. We allow for two linear predictors in the model, but one needs to be "simple" (i.e. only consists of fixed effects). The other is general and defined via the formula. Normally, the zero-inflation probability is simpler (`family="0..."`)

$$\eta_1 = \text{simple} \qquad \eta_2 = \text{formula}$$

but they can also be swapped (`family="0...S"`)

$$\eta_1 = \text{formula} \qquad \eta_2 = \text{simple}$$

Link-function

This is similar to Poisson, Binomial and negative Binomial.

The link-function for the 'simple'-model must be given by argument `link.simple` in the `control.family`-argument. Only link-models without covariates/parameters are currently available. The examples later on shows how this is done.

Hyperparameters

All parameters in the simple model are treated as hyperparameters. The j'th element of η_1 is

$$(\eta_1)_j = \sum_{i=1}^{m} \beta_i x_{ij}$$

for covariates $x_1, \dots,$, where m is maximum 10. An intercept in this model have to be defined manually by adding a constant covariate vector.

New 1inflated models for Poisson

There are two similar models for the Poisson based on the "conditioned to be positive" Poisson $f(y|y > 0)$. These ones are named `"1poisson"` and `"1poissonS"` and are 1-inflated.

Specification

- `family="0poisson"`
- `family="0poissonS"`
- `family="0binomial"`
- `family="0binomialS"`

- `family="0nbinomial"`
- `family="0nbinomialS"`
- `family="1poisson"`
- `family="1poissonS"`
- Required arguments: As for the Poisson, Binomial and negative Binomial (but how these arguments are given, will differ). Optional argument `link.simple`.

  For negative Binomial, only variant 0 and 1 are supported.

Hyperparameter spesification and default values

0poisson

doc `New 0-inflated Poisson`

hyper

- **theta1**
  - **hyperid** `56201`
  - **name** `beta1`
  - **short.name** `beta1`
  - **output.name** `beta1 for 0poisson observations`
  - **output.name.intern** `beta1 for 0poisson observations`
  - **initial** `-4`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `-4 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta2**
  - **hyperid** `56202`
  - **name** `beta2`
  - **short.name** `beta2`
  - **output.name** `beta2 for 0poisson observations`
  - **output.name.intern** `beta2 for 0poisson observations`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `0 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta3**
  - **hyperid** `56203`
  - **name** `beta3`
  - **short.name** `beta3`
  - **output.name** `beta3 for 0poisson observations`
  - **output.name.intern** `beta3 for 0poisson observations`
  - **initial** `0`
  - **fixed** `FALSE`
  - **prior** `normal`
  - **param** `0 10`
  - **to.theta** `function(x) x`
  - **from.theta** `function(x) x`
- **theta4**
  - **hyperid** `56204`
  - **name** `beta4`
  - **short.name** `beta4`
  - **output.name** `beta4 for 0poisson observations`
  - **output.name.intern** `beta4 for 0poisson observations`
  - **initial** `0`
  - **fixed** `FALSE`

prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56205
name beta5
short.name beta5
output.name beta5 for 0poisson observations
output.name.intern beta5 for 0poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56206
name beta6
short.name beta6
output.name beta6 for 0poisson observations
output.name.intern beta6 for 0poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56207
name beta7
short.name beta7
output.name beta7 for 0poisson observations
output.name.intern beta7 for 0poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56208
name beta8
short.name beta8
output.name beta8 for 0poisson observations
output.name.intern beta8 for 0poisson observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta9

hyperid 56209
name beta9
short.name beta9
output.name beta9 for 0poisson observations
output.name.intern beta9 for 0poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta10

hyperid 56210
name beta10
short.name beta10
output.name beta10 for 0poisson observations
output.name.intern beta10 for 0poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56301
name beta1
short.name beta1
output.name beta1 for 0poissonS observations
output.name.intern beta1 for 0poissonS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56302
name beta2
short.name beta2
output.name beta2 for 0poissonS observations
output.name.intern beta2 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56303
name beta3
short.name beta3
output.name beta3 for 0poissonS observations
output.name.intern beta3 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56304
name beta4
short.name beta4
output.name beta4 for 0poissonS observations
output.name.intern beta4 for 0poissonS observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56305
name beta5
short.name beta5
output.name beta5 for 0poissonS observations
output.name.intern beta5 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56306
name beta6
short.name beta6
output.name beta6 for 0poissonS observations
output.name.intern beta6 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56307
name beta7
short.name beta7
output.name beta7 for 0poissonS observations
output.name.intern beta7 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56308
name beta8
short.name beta8
output.name beta8 for 0poissonS observations
output.name.intern beta8 for 0poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56309

name beta9

short.name beta9

output.name beta9 for 0poissonS observations

output.name.intern beta9 for 0poissonS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56310

name beta10

short.name beta10

output.name beta10 for 0poissonS observations

output.name.intern beta10 for 0poissonS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56401
name beta1
short.name beta1
output.name beta1 for 0binomial observations
output.name.intern beta1 for 0binomial observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56402
name beta2
short.name beta2
output.name beta2 for 0binomial observations
output.name.intern beta2 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56403
name beta3
short.name beta3
output.name beta3 for 0binomial observations
output.name.intern beta3 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56404
name beta4
short.name beta4
output.name beta4 for 0binomial observations
output.name.intern beta4 for 0binomial observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56405
name beta5
short.name beta5
output.name beta5 for 0binomial observations
output.name.intern beta5 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56406
name beta6
short.name beta6
output.name beta6 for 0binomial observations
output.name.intern beta6 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56407
name beta7
short.name beta7
output.name beta7 for 0binomial observations
output.name.intern beta7 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56408
name beta8
short.name beta8
output.name beta8 for 0binomial observations
output.name.intern beta8 for 0binomial observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56409

name beta9

short.name beta9

output.name beta9 for 0binomial observations

output.name.intern beta9 for 0binomial observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56410

name beta10

short.name beta10

output.name beta10 for 0binomial observations

output.name.intern beta10 for 0binomial observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56501
name beta1
short.name beta1
output.name beta1 for 0binomialS observations
output.name.intern beta1 for 0binomialS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56502
name beta2
short.name beta2
output.name beta2 for 0binomialS observations
output.name.intern beta2 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56503
name beta3
short.name beta3
output.name beta3 for 0binomialS observations
output.name.intern beta3 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56504
name beta4
short.name beta4
output.name beta4 for 0binomialS observations
output.name.intern beta4 for 0binomialS observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56505
name beta5
short.name beta5
output.name beta5 for 0binomialS observations
output.name.intern beta5 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56506
name beta6
short.name beta6
output.name beta6 for 0binomialS observations
output.name.intern beta6 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56507
name beta7
short.name beta7
output.name beta7 for 0binomialS observations
output.name.intern beta7 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56508
name beta8
short.name beta8
output.name beta8 for 0binomialS observations
output.name.intern beta8 for 0binomialS observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56509

name beta9

short.name beta9

output.name beta9 for 0binomialS observations

output.name.intern beta9 for 0binomialS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56510

name beta10

short.name beta10

output.name beta10 for 0binomialS observations

output.name.intern beta10 for 0binomialS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56421
name beta1
short.name beta1
output.name beta1 for 0nbinomial observations
output.name.intern beta1 for 0nbinomial observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56422
name beta2
short.name beta2
output.name beta2 for 0nbinomial observations
output.name.intern beta2 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56423
name beta3
short.name beta3
output.name beta3 for 0nbinomial observations
output.name.intern beta3 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56424
name beta4
short.name beta4
output.name beta4 for 0nbinomial observations
output.name.intern beta4 for 0nbinomial observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56425
name beta5
short.name beta5
output.name beta5 for 0nbinomial observations
output.name.intern beta5 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56426
name beta6
short.name beta6
output.name beta6 for 0nbinomial observations
output.name.intern beta6 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56427
name beta7
short.name beta7
output.name beta7 for 0nbinomial observations
output.name.intern beta7 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56428
name beta8
short.name beta8
output.name beta8 for 0nbinomial observations
output.name.intern beta8 for 0nbinomial observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56429

name beta9

short.name beta9

output.name beta9 for 0nbinomial observations

output.name.intern beta9 for 0nbinomial observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56430

name beta10

short.name beta10

output.name beta10 for 0nbinomial observations

output.name.intern beta10 for 0nbinomial observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56441
name beta1
short.name beta1
output.name beta1 for 0nbinomialS observations
output.name.intern beta1 for 0nbinomialS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56442
name beta2
short.name beta2
output.name beta2 for 0nbinomialS observations
output.name.intern beta2 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56443
name beta3
short.name beta3
output.name beta3 for 0nbinomialS observations
output.name.intern beta3 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56444
name beta4
short.name beta4
output.name beta4 for 0nbinomialS observations
output.name.intern beta4 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56445
name beta5
short.name beta5
output.name beta5 for 0nbinomialS observations
output.name.intern beta5 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56446
name beta6
short.name beta6
output.name beta6 for 0nbinomialS observations
output.name.intern beta6 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56447
name beta7
short.name beta7
output.name beta7 for 0nbinomialS observations
output.name.intern beta7 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56448
name beta8
short.name beta8
output.name beta8 for 0nbinomialS observations
output.name.intern beta8 for 0nbinomialS observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56449

name beta9

short.name beta9

output.name beta9 for 0nbinomialS observations

output.name.intern beta9 for 0nbinomialS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56450

name beta10

short.name beta10

output.name beta10 for 0nbinomialS observations

output.name.intern beta10 for 0nbinomialS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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1poisson

doc `New 1-inflated Poisson`

hyper

theta1

hyperid `56401`
name `beta1`
short.name `beta1`
output.name `beta1 for 1poisson observations`
output.name.intern `beta1 for 1poisson observations`
initial `-4`
fixed `FALSE`
prior `normal`
param `-4 10`
to.theta `function(x) x`
from.theta `function(x) x`

theta2

hyperid `56402`
name `beta2`
short.name `beta2`
output.name `beta2 for 1poisson observations`
output.name.intern `beta2 for 1poisson observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 10`
to.theta `function(x) x`
from.theta `function(x) x`

theta3

hyperid `56403`
name `beta3`
short.name `beta3`
output.name `beta3 for 1poisson observations`
output.name.intern `beta3 for 1poisson observations`
initial `0`
fixed `FALSE`
prior `normal`
param `0 10`
to.theta `function(x) x`
from.theta `function(x) x`

theta4

hyperid `56404`
name `beta4`
short.name `beta4`
output.name `beta4 for 1poisson observations`
output.name.intern `beta4 for 1poisson observations`
initial `0`
fixed `FALSE`
prior `normal`

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56405
name beta5
short.name beta5
output.name beta5 for 1poisson observations
output.name.intern beta5 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56406
name beta6
short.name beta6
output.name beta6 for 1poisson observations
output.name.intern beta6 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56407
name beta7
short.name beta7
output.name beta7 for 1poisson observations
output.name.intern beta7 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56408
name beta8
short.name beta8
output.name beta8 for 1poisson observations
output.name.intern beta8 for 1poisson observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56409

name beta9

short.name beta9

output.name beta9 for 1poisson observations

output.name.intern beta9 for 1poisson observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56410

name beta10

short.name beta10

output.name beta10 for 1poisson observations

output.name.intern beta10 for 1poisson observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

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hyper

theta1

hyperid 56501
name beta1
short.name beta1
output.name beta1 for 1poissonS observations
output.name.intern beta1 for 1poissonS observations
initial -4
fixed FALSE
prior normal
param -4 10
to.theta function(x) x
from.theta function(x) x

theta2

hyperid 56502
name beta2
short.name beta2
output.name beta2 for 1poissonS observations
output.name.intern beta2 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta3

hyperid 56503
name beta3
short.name beta3
output.name beta3 for 1poissonS observations
output.name.intern beta3 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta4

hyperid 56504
name beta4
short.name beta4
output.name beta4 for 1poissonS observations
output.name.intern beta4 for 1poissonS observations
initial 0
fixed FALSE
prior normal

param 0 10
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 56505
name beta5
short.name beta5
output.name beta5 for 1poissonS observations
output.name.intern beta5 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 56506
name beta6
short.name beta6
output.name beta6 for 1poissonS observations
output.name.intern beta6 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 56507
name beta7
short.name beta7
output.name beta7 for 1poissonS observations
output.name.intern beta7 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 56508
name beta8
short.name beta8
output.name beta8 for 1poissonS observations
output.name.intern beta8 for 1poissonS observations
initial 0
fixed FALSE
prior normal
param 0 10

to.theta function(x) x

from.theta function(x) x

theta9

hyperid 56509

name beta9

short.name beta9

output.name beta9 for 1poissonS observations

output.name.intern beta9 for 1poissonS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

theta10

hyperid 56510

name beta10

short.name beta10

output.name beta10 for 1poissonS observations

output.name.intern beta10 for 1poissonS observations

initial 0

fixed FALSE

prior normal

param 0 10

to.theta function(x) x

from.theta function(x) x

survival FALSE

discrete TRUE

link default logit loga cauchit probit cloglog ccloglog loglog log sslogit logitoffset quantile pquanti

link.simple default log

pdf 0inflated

Example: 0Poisson

```
sim.poisson <- function(prob, m)
{
    stopifnot(length(prob) == length(m) && length(prob) > 0)
    n <- length(m)
    y <- numeric(n)
    event <- (runif(n) < prob)
    idx.zero <- which(event)
    idx.non.zero <- which(!event)
    y[idx.zero] <- 0
    y[idx.non.zero] <- rpois(length(idx.non.zero), lambda = m[idx.non.zero])
    return (y)
}

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

n <- 1000
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
E <- runif(n, min = 0.8, max = 1/0.8)

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
m <- E*exp(eta2)

ok <- FALSE
while(!ok) {
    y <- sim.poisson(prob, m)
    ok <- !all(y == 0)
}

## head(data.frame(y, E, x, z, xx, zz))

r <- inla(
    inla.mdata(cbind(y, E), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
    family = "0poisson",
    data = data.frame(y, E, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.family = list(link.simple = link.simple,
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
                                       beta3 = list(param = c(0, 1)),
                                       beta4 = list(param = c(0, 1)),
                                       beta5 = list(param = c(0, 1))))))

rr <- inla(
```

```
    inla.mdata(cbind(y, E), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
    family = "0poissonS",
    data = data.frame(y, E, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    ## in this case we need to define link.simple as the main link
    control.family = list(control.link = list(model = link.simple),
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
                                       beta3 = list(param = c(0, 1)),
                                       beta4 = list(param = c(0, 1)),
                                       beta5 = list(param = c(0, 1)))))
summary(r)
summary(rr)

res <- cbind("beta" = beta,
             "0poisson" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
             "0poissonS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
             diff = (res[, 2]-beta),
             diffS = (res[, 3]-beta),
             "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
             "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".poisson"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))
```

Example: 0Binomial

```
sim.binomial <- function(prob, p, size)
{
    ## - prob=zero-inflation-prob
    ## - binomial(size, p)
    stopifnot(length(prob) == length(p) && length(prob) == length(size)
              && length(prob) > 0)
    n <- length(prob)
    y <- numeric(n)
    event <- (runif(n) < prob)
    idx.zero <- which(event)
    idx.non.zero <- which(!event)
    y[idx.zero] <- 0
    y[idx.non.zero] <- rbinom(length(idx.non.zero),
                              size = size[idx.non.zero],
                              prob = p[idx.non.zero])
    return (y)
}

n <- 1000
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
Ntrials <- sample(1:10, n, replace = TRUE)

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
p <- 1/(1 + exp(-eta2))

ok <- FALSE
while(!ok) {
    y <- sim.binomial(prob, p, Ntrials)
    ok <- !all(y == 0)
}

## head(data.frame(y, Ntrials, x, z, xx, zz))

r <- inla(
    inla.mdata(cbind(y, Ntrials), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
    family = "0binomial",
    data = data.frame(y, Ntrials, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.family = list(link.simple = link.simple,
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
```

```
                                    beta3 = list(param = c(0, 1)),
                                    beta4 = list(param = c(0, 1)),
                                    beta5 = list(param = c(0, 1)))))

rr <- inla(
    inla.mdata(cbind(y, Ntrials), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
    family = "0binomialS",
    data = data.frame(y, Ntrials, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    ## in this case we need to define link.simple as the main link
    control.family = list(control.link = list(model = link.simple),
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
                                       beta3 = list(param = c(0, 1)),
                                       beta4 = list(param = c(0, 1)),
                                       beta5 = list(param = c(0, 1)))))

summary(r)
summary(rr)

res <- cbind("beta" = beta,
             "0binomial" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
             "0binomialS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
             diff = (res[, 2]-beta),
             diffS = (res[, 3]-beta),
             "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
             "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".binomial"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))
```

Example: 1Poisson

```
sim.1poisson <- function(prob, lambda) {
    stopifnot(length(prob) == length(lambda) && length(prob) > 0)
    n <- length(lambda)
    y <- numeric(n)
    event <- (runif(n) < prob)
    idx.one <- which(event)
    idx.non.one <- which(!event)
    y[idx.one] <- 1
    for(i in idx.non.one) {
        y[i] <- 0
        while(y[i] == 0) {
            yy <- rpois(20, lambda = lambda[i])
            if (any(yy > 0)) {
                y[i] <- yy[min(which(yy > 0))]
            }
        }
    }
    return (y)
}

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit
## link.simple <- "probit"
## inv.link <- inla.link.invprobit
## link.simple <- "cloglog"
## inv.link <- inla.link.invcloglog

n <- 10^4
z <- rnorm(n, sd = 0.3)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.3)
zz <- rnorm(n, sd = 0.2)
E <- runif(n, min = 0.1, max = 100)

beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xx * zz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * x * z
prob <- inv.link(eta1)
lambda <- E*exp(eta2)

y <- sim.1poisson(prob, lambda)

## head(data.frame(y, E, x, z, xx, zz))

r <- inla(
    inla.mdata(cbind(y, E), cbind(1, x, z, x*z)) ~ 1 + xx + zz + xx*zz,
    family = "1poisson",
    data = data.frame(y, E, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.family = list(link.simple = link.simple,
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
                                       beta3 = list(param = c(0, 1)),
                                       beta4 = list(param = c(0, 1)),
```

```
                                    beta5 = list(param = c(0, 1)))))

rr <- inla(
    inla.mdata(cbind(y, E), cbind(1, xx, zz, xx*zz)) ~ 1 + x + z + x*z,
    family = "1poissonS",
    data = data.frame(y, E, x, z, xx, zz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.inla = list(cmin = 0),
    ## in this case we need to define link.simple as the main link
    control.family = list(control.link = list(model = link.simple),
                          hyper = list(beta1 = list(param = c(0, 1)),
                                       beta2 = list(param = c(0, 1)),
                                       beta3 = list(param = c(0, 1)),
                                       beta4 = list(param = c(0, 1)),
                                       beta5 = list(param = c(0, 1)))))
summary(r)
summary(rr)

res <- cbind("beta" = beta,
             "1poisson" = c(r$summary.fixed$mean, r$summary.hyperpar$mean),
             "1poissonS" = c(rr$summary.hyperpar$mean, rr$summary.fixed$mean))
res <- cbind(res,
             diff = (res[, 2]-beta),
             diffS = (res[, 3]-beta),
             "diff/sd" = (res[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd),
             "diffS/sd" = (res[, 3]-beta) / c(rr$summary.hyperpar$sd, rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".poisson"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 2, res))
```

Example: 0nBinomial

```
sim.nbinom <- function(prob, m, size)
{
    stopifnot(length(prob) == length(m) &&
              length(size) == length(m) &&
              length(prob) > 0)
    n <- length(m)
    y <- numeric(n)
    event <- (runif(n) < prob)
    idx.zero <- which(event)
    idx.non.zero <- which(!event)
    y[idx.zero] <- 0
    y[idx.non.zero] <- rnbinom(length(idx.non.zero),
                               mu = m[idx.non.zero],
                               size = size[idx.non.zero])
    return (y)
}

## chose link-function to use for the zero-inflation probability
link.simple <- "logit"
inv.link <- inla.link.invlogit

variant <- 0
n <- 10^4
z <- rnorm(n, sd = 0.2)
x <- rnorm(n, sd = 0.2)
xx <- rnorm(n, sd = 0.2)
zz <- rnorm(n, sd = 0.2)
E <- runif(n, min = 0.5, max = 1/0.5)
if (variant == 1) {
    size = 3 * E
} else {
    size <- rep(3, n)
}

xxzz <- 0.2 * scale(xx * zz)
xz <- 0.2 * scale(x*z)
beta <- c(1, 1.1, 2.1, 0, -2, 1.2, 2.2, 0)
eta2 <- beta[1] + beta[2] * xx + beta[3] * zz + beta[4] * xxzz
eta1 <- beta[5] + beta[6] * x + beta[7] * z + beta[8] * xz
prob <- inv.link(eta1)
m <- E*exp(eta2)

ok <- FALSE
while(!ok) {
    y <- sim.nbinom(prob, m, size)
    ok <- !all(y == 0)
}

r <- inla(
    inla.mdata(cbind(y, E), cbind(1, x, z, xz)) ~ 1 + xx + zz + xxzz,
    family = "0nbinomial",
    data = data.frame(y, E, x, z, xz, xx, zz, xxzz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.family = list(link.simple = link.simple, variant = variant,
                          hyper = list(beta1 = list(param = c(0, 1)),
```

```
                                    beta2 = list(param = c(0, 1)),
                                    beta3 = list(param = c(0, 1)),
                                    beta4 = list(param = c(0, 1)))))
##r <- inla.rerun(r)
summary(r)

rr <- inla(
    inla.mdata(cbind(y, E), cbind(1, xx, zz, xxzz)) ~ 1 + x + z + xz,
    family = "0nbinomialS",
    data = data.frame(y, E, x, z, xz, xx, zz, xxzz),
    control.fixed = list(prec = 1, prec.intercept = 1),
    control.compute = list(cpo = TRUE),
    control.inla = list(cmin = 0),
    ## in this case we need to define link.simple as the main link
    control.family = list(control.link = list(model = link.simple),
                          hyper = list(beta1 = list(param = c(0, 1)),
                                    beta2 = list(param = c(0, 1)),
                                    beta3 = list(param = c(0, 1)),
                                    beta4 = list(param = c(0, 1)))))
##rr <- inla.rerun(rr)

summary(r)
summary(rr)

ress <- cbind("beta" = beta,
              "0nbin" = c(r$summary.fixed$mean, r$summary.hyperpar$mean[-1]),
              "0nbinS" = c(rr$summary.hyperpar$mean[-1], rr$summary.fixed$mean))
res <- cbind(ress,
              diff = (ress[, 2]-beta),
              diffS = (ress[, 3]-beta),
              "sd" = c(r$summary.fixed$sd, r$summary.hyperpar$sd[-1]),
              "sdS" = c(rr$summary.hyperpar$sd[-1], rr$summary.fixed$sd),
              "diff/sd" = (ress[, 2]-beta) / c(r$summary.fixed$sd, r$summary.hyperpar$sd[-1]),
              "diffS/sd" = (ress[, 3]-beta) / c(rr$summary.hyperpar$sd[-1], rr$summary.fixed$sd))
mm <- nrow(res) %/% 2
rownames(res) <- c(paste0("beta", 1:mm, ".nbin"), paste0("beta", 1:mm, ".prob"))
print(round(dig = 3, res))

round(dig = 3, rbind(r$summary.hyperpar[1,c(1, 2, 6)],
                     rr$summary.hyperpar[1,c(1, 2, 6)]))
plot(r$cpo$pit, rr$cpo$pit)
abline(a=0,b=1,lwd=3,col="blue")
coef(lm(r$cpo$pit ~ rr$cpo$pit))
```